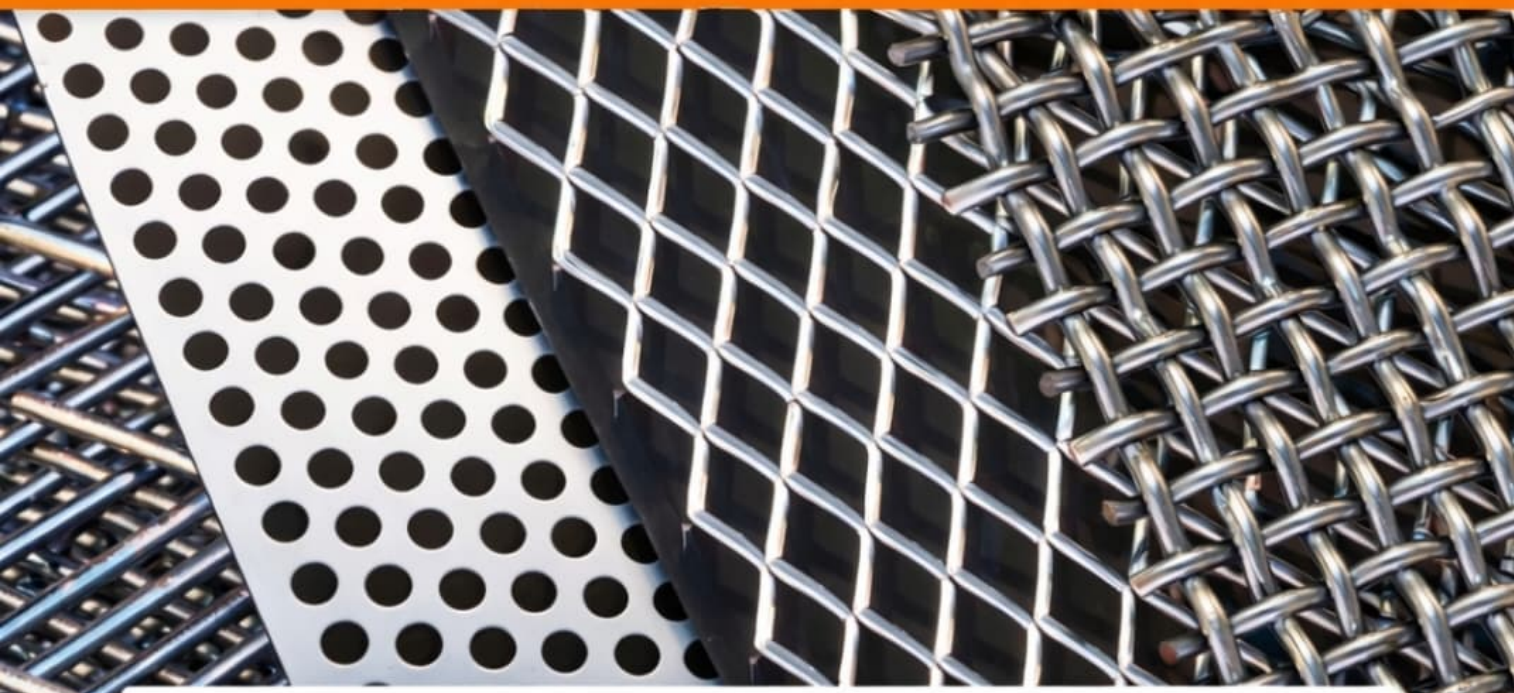
National Steel



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NATIONAL STEEL

Why Us

Experience and Expertise: With over three decades of industry experience, we are leading manufacturer and supplier of high-quality wire mesh products. Our team of skilled professionals possesses extensive expertise in wire weaving and perforation that meet exact specifications of customers.

Wide Range of Products: At National Steel, we take pride in ffering an extensive range of wire mesh products to cater to diverse industrial needs, from filtration an screening to security and security ad construction.

Quality and Customer Satisfaction: Our commitment to delivering unmatched quality is the cornerstone of our success. We adhere to strict quality control cnea-measures throughout the manufacturing process, using premium-grade materials & advanced technologies to produce durable and reliable wire mesh products.

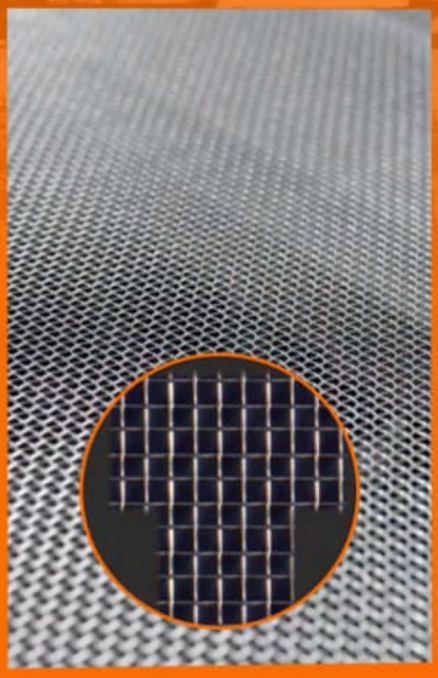
- Woven Wire Mesh
- Crimped Wire Mesh
- Welded Mesh
- Vibrating Screen Mesh
- Perforated Sheets
- Expanded Metal
- Chainlink Fencing

Woven mesh

Woven wire mesh is a versatile metal product that is made by weaving wire strands together. It is widely used in various industries, including construction, mining, agriculture, and manufacturing, and is known for its strength and durability. Wire mesh is used for filtering, sieving, fencing, security screens, and insect screening. It can be made from different materials, including steel, stainless steel, and brass, and is available in different sizes and styles.

Product Range

- 1 Mesh to 2300 Mesh
- Opening from 0.002MM (2 Micron) to 22 MM
- Material – SS304, SS316, Brass, Phosphor Bronze, Copper, Monel, Aluminum, Galvanised Iron
- Order Type – Ready Stock, Make to Order



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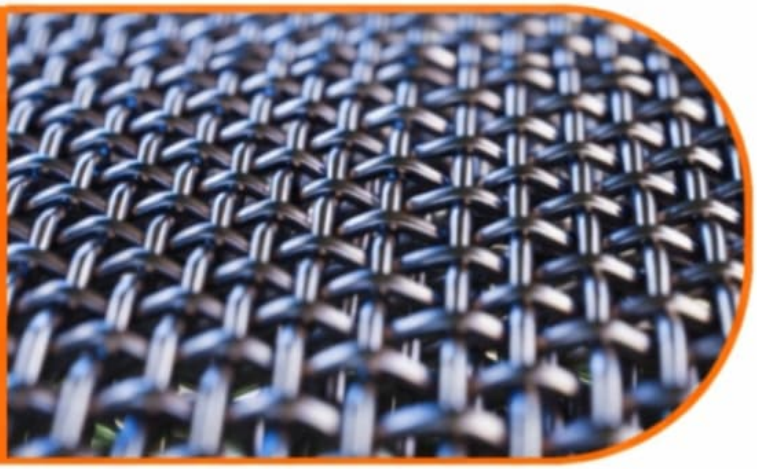
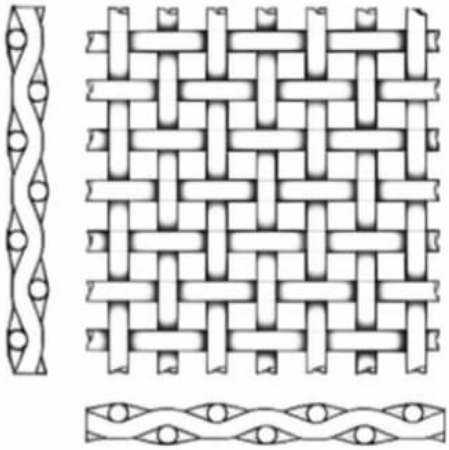
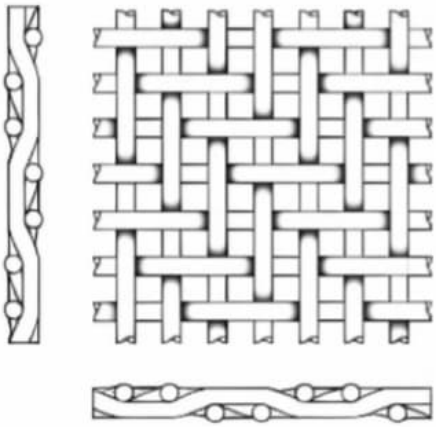
Types of Weave

Wire mesh can be woven in several different ways to different patterns and densities. Here are some of the most common types of weave:

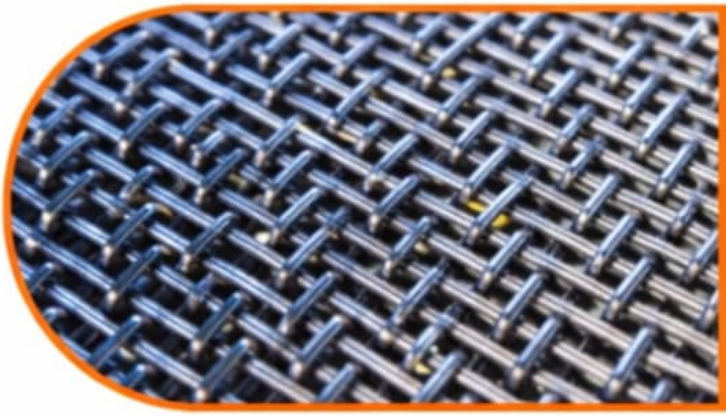
Plain Weave



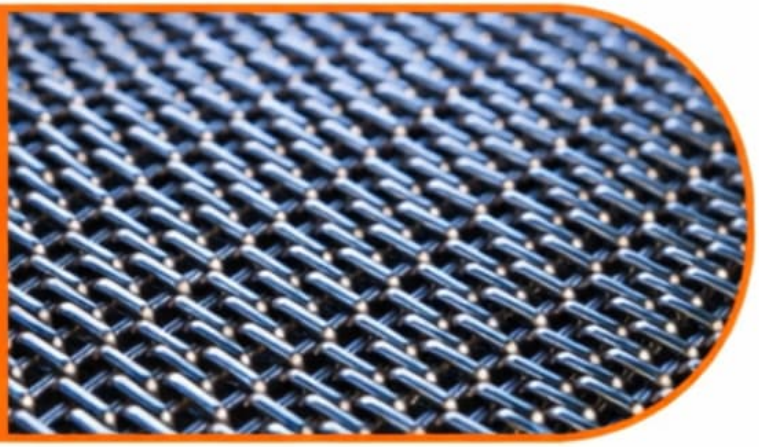


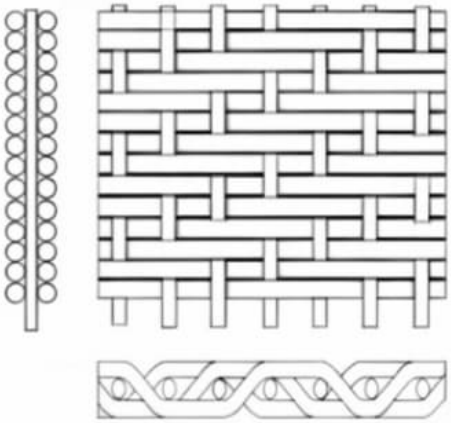


Twill Weave



Dutch Weave





Woven mesh

Plain Weave Wire Mesh Technical

Stated below are the technicals of the Plain Weave wire Mesh. These items are mostlty available in stock in standard Width sizes. Any special requirement can be manufactured to order. The standerd material for these meshes are SS 304,SS 304L, SS 316, SS 316L & SS 310

Mesh	Diameter of Wire		Opening in MM	Approx Open Area %
	SWG	MM		
1	-	3	22.40	77%
1	-	2.5	22.90	81%
2	-	3	9.70	58%
2	-	2.5	10.20	64%
2	14	2	10.70	70%
3	-	2.5	5.96	49%
3	14	2	6.46	58%
3	-	1.5	6.96	67%
4	14	2	4.35	46%
4	-	1.5	4.85	58%
4	18	1.2	5.15	65%
4	20	0.9	5.45	73%
5	14	2	3.08	36%
5	-	1.5	3.58	49%
5	18	1.2	3.88	58%
6	-	1.5	2.73	41%
6	18	1.2	3.03	51%
6	20	0.9	3.33	62%
8	-	1.5	1.67	27%
8	18	1.2	1.97	38%
8	19	1	2.17	46%
8	20	0.9	2.27	51%
8	21	0.8	2.37	55%
8	22	0.7	2.47	60%
8	23	0.6	2.57	65%
8	24	0.55	2.62	68%
8	25	0.5	2.67	70%
10	20	0.9	1.64	41%
10	21	0.8	1.74	46%
10	22	0.7	1.84	52%
10	23	0.6	1.94	58%
10	24	0.55	1.99	61%
10	25	0.5	2.04	64%
12	22	0.7	1.41	44%

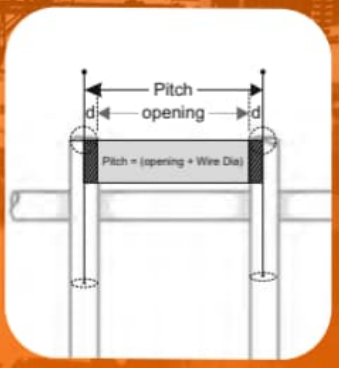
Mesh	Diameter of Wire		Opening in MM	Approx Open Area %
	SWG	MM		
12	23	0.6	1.51	51%
12	24	0.55	1.56	54%
12	25	0.5	1.61	58%
12	26	0.45	1.66	62%
14	24	0.55	1.26	48%
14	25	0.5	1.31	52%
14	26	0.45	1.36	56%
14	27	0.4	1.41	60%
16	23	0.6	0.98	38%
16	24	0.55	1.03	42%
16	25	0.5	1.08	46%
16	26	0.45	1.13	51%
16	27	0.4	1.18	55%
16	28	0.37	1.21	58%
16	30	0.3	1.28	65%
16	32	0.27	1.31	68%
16	33	0.25	1.33	70%
16	34	0.22	1.36	74%
18	24	0.55	0.86	37%
18	25	0.5	0.91	41%
18	26	0.45	0.96	46%
18	27	0.4	1.01	51%
18	28	0.37	1.04	54%
18	30	0.3	1.11	62%
18	32	0.27	1.14	65%
18	33	0.25	1.16	67%
20	24	0.55	0.72	32%
20	25	0.5	0.77	36%
20	26	0.45	0.82	41%
20	27	0.4	0.87	46%
20	28	0.37	0.90	50%
20	30	0.3	0.97	58%
20	32	0.27	1.00	62%
20	33	0.25	1.02	64%
20	36	0.19	1.08	72%

Mesh	Diameter of Wire		Opening in MM	Approx Open Area %
	SWG	MM		
24	27	0.4	0.65	38%
24	28	0.37	0.68	42%
24	30	0.3	0.75	51%
24	32	0.27	0.78	55%
30	28	0.37	0.47	31%
30	30	0.3	0.54	41%
30	32	0.27	0.57	46%
30	33	0.25	0.59	49%
30	34	0.22	0.62	54%
30	36	0.19	0.65	60%
36	33	0.25	0.45	41%
36	34	0.22	0.48	47%
36	36	0.19	0.51	53%
40	32	0.27	0.36	33%
40	33	0.25	0.38	36%
40	34	0.22	0.41	42%
40	36	0.19	0.44	49%
50	34	0.22	0.28	32%
50	36	0.19	0.31	39%
50	38	0.15	0.35	49%
60	36	0.19	0.23	30%
60	38	0.15	0.27	41%
80	40	0.12	0.19	38%
80	42	0.1	0.21	46%
100	42	0.1	0.15	36%
100	44	0.08	0.17	46%
120	44	0.08	0.13	38%
150	45	0.07	0.09	34%
150	46	0.06	0.10	41%
200	47	0.05	0.077	36%
250	48	0.04	0.061	36%
300	48	0.04	0.044	27%
325	-	0.035	0.043	30%
400	49	0.03	0.033	27%
500	50	0.025	0.025	25%

Welded mesh

Our offering

Specification	Description
Material	Stainless steel, Galvanized Steel, Mild Steel, etc.
Wire Diameter	Typically ranges from 1mm to 6mm or more
Mesh Size	Can vary from 15 mm to 6 inches
Width	Typically ranges from 1 meter to 2 meters
Length	Can be customized according to customer requirements
Tensile Strength	Depends on the material and wire diameter used
Corrosion Resistance	High for Stainless Steel and Galvanized Steel options
Applications	Reinforcing Concrete, Fencing, guards, Enclosures, etc.

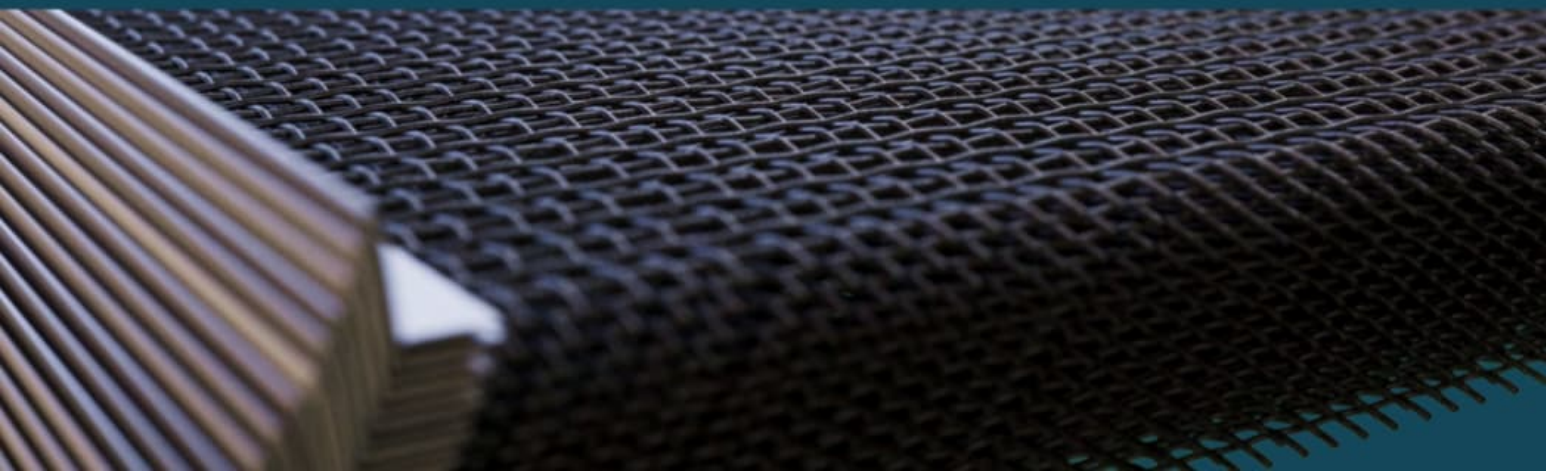


Vibrating Screen Mesh

Vibrating screen wire mesh is a type of screen that is used for filtering or separating different materials based on their size. It is made from high-quality spring steel (high carbon) material that provides superior strength, durability, and resistance to abrasion.

The vibrating screen wire mesh works by using a vibrating motor to create vibrations that cause the screen mesh to vibrate, allowing materials to pass through the openings in the mesh based on their size. The vibrating motor is attached to the screen frame, and as it rotates, it creates vibrations that cause the material on the screen to move and separate into different sizes.

Vibrating screen wire mesh is commonly used in industries such as mining, quarrying, construction, and others for the efficient screening of various materials. It is available in a range of sizes and openings to suit different requirements and can be customized to fit specific dimensions as needed. It is also easy to install and maintain, making it a cost-effective solution for your screening needs.



PRODUCTS

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Highlights

High-quality material: Vibrating screen wire mesh is maile ed from high-quality spring steel (high carbon) material, which is known for its durability and strength. Material sourced from TATA or USHA MARTIN specially engineered for crimping and weaving.

Versatile design: Vibrating screen wire mesh can uused in a variety of applications, such as in mining, quarrying, and construction, among others.

Precise sizing: Vibrating screen wire mesh is avvailable in a rangers of sizes and openings to suit different requirements and can be customized to fit specific dimensions as needed. Can be made to sizes of various OEM like Metso, Puzzolana, Sandvik, Asia, Propel, Hailstone

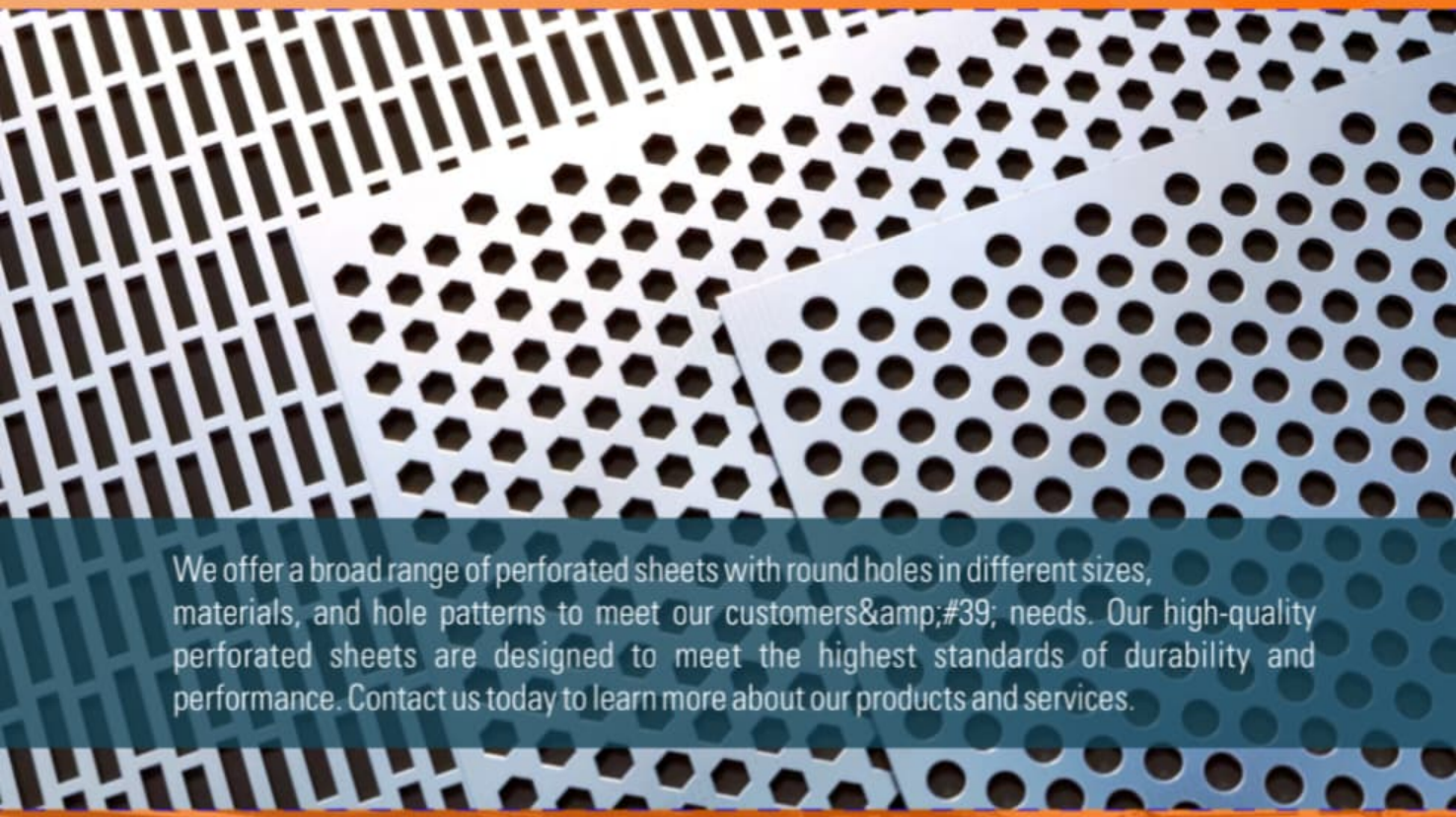
Efficient screening: Vibrating screen wire mesh is designed to provide efficient screening of various materials, including aggregates, minerals, and other substances.

Easy installation: Vibrating screen wire mesh is typically easy to instal and can be attached to the screen frame using clamps or other fastening mechanisms.

Aperture	Wire Dia	Aperture	Wire Dia	Aperture	Wire Dia	Aperture	Wire Dia
2	0.9	12	3	20	8	30	6-10
2.5	1	16	4	22	6	30	6-10
3	1,6	12	5	22	6	30	6-10
4	2	16	5	22	6	30	6-10
5	2.5	12	6	23	8	32	6-10
6	3	18	6	24	8	32	6-10
7	3.0	16	6	26	8	34	8-10
7	3	18	6	28	8	42	6-10
8	4	16	6	30	8	42	8-10
9	5	18	6	30	8	44	8-10
9	5	18	6	32	10	50	8-10
9	6	18	6	30	6	60	8-10
10	5	18	6	35	8	50	8-10
10	6	18	6	35	8	65	8-10
10	7	18	6	38	8	70	8-10
10	8	18	6	40	8	90	8-10
10	8	18	6	45	8	55	8-10
10	8	18	6	50	8	54	8-10
10	8	18	6	56	8	60	8-10
10	10	18	6	70	8	80	8-10

* Customized sizes can also be available on demand

Perforated Sheets



We offer a broad range of perforated sheets with round holes in different sizes, materials, and hole patterns to meet our customers' needs. Our high-quality perforated sheets are designed to meet the highest standards of durability and performance. Contact us today to learn more about our products and services.

Perforated sheets can be used for filtration, ventilation, decoration, and many other purposes. The holes can be sized to meet specific requirements and can range from very small to very large. This makes them ideal for a wide range of applications, including air and liquid filtration, noise reduction, and heat dissipation.
Moreover, perforated sheets with round holes can be made from a variety of materials, such as stainless steel, aluminum, brass, and mild steel. This allows for customization of the sheets to suit specific requirements of the industry.

Applications

- Fabrication
- Engineering
- Automobiles
- Food
- Agriculture
- Decorative
- Sound Absorption

Materials

- Mild Steel
  - a.HR (Hot Rolled)
  - b.CR (Cold Rolled)
- Stainless Steel
  - a.SS 304
  - b.SS 316
  - c.SS 430
- Galvanised Iron (GI)
- Aluminium
- Brass

Standard sizes

- 1.25 X 2.5 Meter
- 1.5 X 3 Meter
- 1 X 2 Meter

Custom Sizes on Request

Thickness

- 0.35MM to 12 MM

Hole shape

- Round
- Square
- Slot
- Hex

Hole diameter

- 0.5mm to 100mm

Expanded Metal

Expanded metal is a versatile and durable product that can be used in a wide range of applications, from industrial and commercial to decorative and architectural. To help you choose the right expanded metal product for your needs, we've put together this technical fact sheet that covers some of the key details you should know about expanded metal.

Material: Expanded metal can be made from a variety of materials, including steel, aluminum, stainless steel, and other metals. The material used can affect the strength, durability, and corrosion resistance of the expanded metal.

Thickness: The thickness of the metal sheet before it was expanded is an important factor to consider, as it can affect the strength and rigidity of the expanded metal. Thickness is usually measured in gauge or mm.

SWD and LWD: The diamond-shaped openings in the expanded metal sheet are formed by stretching and cutting the metal strands in a specific pattern. The width and length of the strands can affect the size and shape of the diamond openings.

Strand thickness: The thickness of the strands can also affect the strength and durability of the expanded metal, as well as the size of the diamond openings.

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Highlights

Open area: The percentage of open area in the expanded metal sheet is an important factor to consider for certain applications where air or light flow is required. Open area can be calculated based on the size and spacing of the diamond openings.

Standard sizes: Expanded metal sheets are available in a range of standard sizes, including width and length dimensions. Knowing the standard sizes available can help you determine the most cost-effective option for your needs..

Applications: Expanded metal can be used for a variety of applications, such as fencing, grating, walkways, filters, and decorative purposes. The application will determine the material, thickness, strand width and length, and other factors required for the expanded metal.

Customization: If you need expanded metal in a size, shape, or finish that isn't available as a standard option, customization may be possible. Contact us aout custom orders.

In summary, expanded metal is a versatile and durable product that can be customized to suit a variety of applications. By understanding the material, thickness, strand width and length, open area, surface finish, applications, customization options, and standards compliance, you can choose the right expanded metal product for your needs.

Chainlink Fencing

Fencing, Whether you're looking to secure your property, keep pets in, or simply add a decorative touch to your yard, our chainlink fencing is a reliable and affordable option. Made from high-quality materials, our fencing is designed to withstand the elements and keep your property safe and secure for years to come.

Product Features:

Durability: Our chainlink fencing is made from galvanized steel, which is rust-resistant and durable enough to withstand harsh weather conditions.
Security: With its sturdy construction and tight weave, our chainlink fencing provides a secure barrier to keep unwanted visitors out.
Customization: We offer a range of fence heights, mesh sizes, and colors to suit your specific needs and preferences.

Product Benefits:

Affordability: Chainlink fencing is one of the most cost-effective fencing options on the market, making it an excellent choice for those on a budget.
Low Maintenance: Unlike other fencing materials, chainlink requires little to no maintenance. Simply hose it down occasionally to keep it looking its best.
Versatility: Chainlink fencing can be used for a variety of applications, from residential to commercial, and is suitable for both small and large properties.

If you're in the market for a reliable and affordable fencing solution, our chainlink fencing is an excellent choice. With its durability, security, and customization options, it's sure to meet all your needs and exceed your expectations. Order now and enjoy the peace of mind that comes with knowing your property is safe and secure.